Facet capacity and pentagon excess in compact right-angled hyperbolic four-polytopes

Coxeter Polytopes Frontier Project

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Abstract

Let P be a compact right-angled hyperbolic four-polytope, with f facets and total ridge excess W above pentagons. A facet with V vertices forces $f \geq 2V+5$ and $W \geq \lceil 3(V-20)/4 \rceil$. Consequently every facet has at most 46 vertices at or below the volume of the regular right-angled 120-cell, and at most 96 strictly below its double. We also prove $W = 0$ or $W \geq 17$. The zero case determines the whole 120-cell; the positive case uses regular-pentagon edge lengths and an explicit small-graph argument. If a positive facet has only two nonpentagonal neighbors, then $W \geq 20$. The capacity proof and its three-dimensional equality case are given in full. The later metric arguments state their three-dimensional uniqueness and dodecahedron-classification inputs explicitly. These are constraints on possible smaller-volume polytopes, not a universal minimum-volume or second-volume theorem.

1 Statement and scope

Throughout, a polytope is convex and full-dimensional. A right-angled hyperbolic polytope has dihedral angle $\pi/2$ at every ridge. All polytopes in this paper are compact. In particular every edge has two finite endpoints, and every two-face is a compact right-angled polygon. If $P \subset \mathbb{H}^4$, write (v, e, r, f) for its numbers of vertices, edges, ridges and facets.

Theorem 1. *Let F be a facet of a compact right-angled hyperbolic four-polytope P. If F has V vertices, then*

$$f \geq 2V + 5. \tag{1}$$

Define the pentagon excess by

$$W = \sum_{R \text{ a ridge of } P} \big(k(R) - 5\big),$$

where $k(R)$ is the number of sides of R. We will prove

$$W = v - 5f, \qquad \mathrm{Vol}(P) = \frac{\pi^2}{12}(f + W + 16). \tag{2}$$

For a facet F, put

$$d(F) = \sum_{R \subset F} \big(k(R) - 5\big).$$

If it has V vertices, then

$$d(F) = \frac{V - 20}{2}, \qquad \sum_F d(F) = 2W. \tag{3}$$

Theorem 2. *Every facet with V vertices satisfies*

$$W \geq \left\lceil \frac{3(V-20)}{4} \right\rceil, \qquad f + W \geq 2V + 5 + \left\lceil \frac{3(V-20)}{4} \right\rceil \geq \frac{11V}{4} - 10. \tag{4}$$

Corollary 3. *The following local size restrictions hold.*

1. *If* $\mathrm{Vol}(P) \le 34\pi^2/3$*, every facet has at most* 46 *vertices.*
2. *If* $\mathrm{Vol}(P) < 68\pi^2/3$*, every facet has at most* 96 *vertices.*

Theorem 4. *The pentagon excess satisfies* $W = 0$ *or* $W \ge 17$*. Moreover* $W = 0$ *if and only if* P *is isometric to the regular right-angled* 120*-cell.*

The comparison values are the volumes of that 120-cell and its facet double. A distinct polytope at or below the first value must have $W \ge 17$ as well as the local cutoff 46. The inequality $f + W \ge 120$ is still not proved. Neither universal minimality, a second volume gap, nor decomposition into regular 120-cells is assumed or deduced.

The geometric input is the usual acute-chamber wall property: distinct outward unit normals of an acute-angled hyperbolic polytope have nonpositive scalar products. We use the standard Lorentz model and the ordinary dual cell decomposition of the boundary of a simple convex polytope. These are the acute-angle facts discussed in [2, pp. 63–64 and Proposition 4.1], with the general geometric background in [1]. We prove below the stronger compact normal gap, the intersection and distinctness statements used in the count, and the needed planar closing argument. The volume discussion also uses Chern–Gauss–Bonnet for a compact hyperbolic reflection orbifold; its resulting face formula agrees with [3, Proposition 3.2(d)]. The capacity and neighboring-excess inequalities use no small-polyhedron classification. The positive-excess gap and zero-excess recognition use two further inputs, isolated in Section 6: a compact right-angled three-polyhedron with 12 faces is the regular dodecahedron, and the compact nonobtuse three-dimensional Andreev uniqueness theorem. No classification or volume-minimum theorem in dimension four is an input. Standard convex-polytope facts used below are Euler's formula, the dual boundary sphere, and connectedness of facet adjacency.

The perpendicular-edge mechanism has an explicit antecedent in Kellerhals–Perren's proof of Proposition 3.2(b), p. 1311 [3]: the outgoing edges of a facet with N vertices yield N distinct further facets. They use this to prove $f^2 > 4v$. Our count strengthens that mechanism by controlling the extra facets needed around its neighbors. The three-dimensional excess identity is already in Inoue's Lemma 3.3 [4]. Neither that identity nor the opposite-facet mechanism is claimed as new here.

2 Normal geometry and perpendicular edges

Use the hyperboloid model in $\mathbb{R}^{n,1}$, with scalar product of signature $(-,+,\dots,+)$. A point of $\mathbb{H}^n$ is a future unit timelike vector. An outward unit spacelike normal a defines the half-space $\langle x, a\rangle \le 0$.

Lemma 5. *Distinct wall normals of a compact right-angled polytope have product either* 0 *or strictly less than* -1*. A mutually orthogonal collection of wall normals determines an actual common face.*

Proof. Products are nonpositive by the acute-chamber wall property. Suppose two normals a, b have $g = \langle a, b\rangle \in (-1, 0)$. Their Gram matrix

$$G = \begin{pmatrix} 1 & g \\ g & 1 \end{pmatrix}$$

is positive definite and G^{-1} has nonnegative entries. Let p be a strict future timelike interior point and set $\beta = (\langle p, a\rangle, \langle p, b\rangle)^T < 0$. The orthogonal projection

$$u = p - [a\ b]G^{-1}\beta$$

adds nonnegative multiples of the outward normals to p. Thus it strictly satisfies every other wall inequality. It satisfies the two selected equalities, and

$$u^2 = p^2 - \beta^T G^{-1} \beta < 0, \qquad \langle p, u \rangle = u^2 < 0.$$

It is therefore future timelike and gives an actual ridge, whose angle is not $\pi/2$, a contradiction.

If $\langle a, b \rangle = -1$, the nonzero vector $a + b$ is null. It has negative product with p, hence is future. Its products with all other wall normals are nonpositive. It would be a feasible ideal point, contrary to compactness. The sum cannot be zero, because p has negative product with both summands. This proves the gap.

For mutually orthogonal normals $a_1, \ldots, a_j$, the same projection is particularly simple:

$$u = p - \sum_{i=1}^{j} \langle p, a_i \rangle a_i.$$

It is future timelike, has zero product with each selected normal and strict negative product with every other wall normal. Hence their intersection is an actual codimension-j face. □

At any finite vertex, all its incident normals lie in a positive definite space. Lemma 5 makes them mutually orthogonal. There are exactly n of them, so the polytope is simple. A facet inherits right angles: its neighboring wall normals are orthogonal to its own normal, and serve as its intrinsic outward normals. The same applies to faces of lower dimension. A compact right-angled k-gon has area $(k-4)\pi/2 > 0$, so $k \geq 5$.

Lemma 6. *Fix a facet A with normal a, and a facet disjoint from A with normal z. Any two common neighbors of these facets meet each other.*

Proof. We have $\langle a, z \rangle < -1$. The plane spanned by a, z is Lorentzian, so its orthogonal complement is positive definite. The normals b, c of two common neighbors belong to this complement. In a positive definite space their product cannot be below -1. Product -1 would imply $b = -c$, incompatible with a strict point of both negative half-spaces. Lemma 5 therefore gives $\langle b, c \rangle = 0$, and the facets meet. □

We call the set of neighbors of A meeting another facet its *footprint on the neighbors of A*. Lemma 6 says that a disjoint facet's footprint is a clique. This is the only meaning of footprint used below.

Lemma 7. *Let A be a facet of a compact right-angled polytope in dimension three or four. For every vertex x of A, the edge leaving A is perpendicular to A and ends on a facet R_x disjoint from A. All the R_x are distinct. Moreover, R_x meets a neighbor B of A if and only if x belongs to B.*

An actual facet disjoint from A meeting all the neighbors of A at x must be R_x.

Proof. Let a be the outward normal of A. Regard x as a vector in $a^\perp$. Simplicity gives exactly one edge at x not contained in A. All its other defining walls are orthogonal to a, so its initial direction is $-a$. If its length is $L_x > 0$, the other endpoint is

$$y = \cosh(L_x)x - \sinh(L_x)a.$$

The new normal there is forced to be

$$r_x = \sinh(L_x)x - \cosh(L_x)a. \tag{5}$$

It is orthogonal to the walls defining the edge, has norm one and satisfies $\langle y, r_x \rangle = 0$, $\langle x, r_x \rangle < 0$. Its product with a is $-\cosh(L_x) < -1$, so R_x is disjoint from A.

For a neighbor B with normal b,

$$\langle r_x, b\rangle = \sinh(L_x)\langle x, b\rangle.$$

This is zero exactly when $x \in B$; the actual-intersection part of Lemma 5 gives the stated incidence equivalence. If $r_x = r_y$, projection to $a^\perp$ gives $\sinh(L_x)x = \sinh(L_y)y$, hence $x = y$.

Finally a normal orthogonal to all the neighbors defining x lies in $\operatorname{span}(x, a)$. For an actual facet disjoint from A its orientation and its product with x give the form $r = \sinh(L)x - \cosh(L)a$, with $L > 0$. Two distinct normals of this form have product $\cosh(L - L_x) > 0$, contrary to acuteness. Thus $r = r_x$. □

The final assertion is useful: a more distant wall on the same perpendicular line cannot be counted as a second actual facet.

3 Closing the neighborhood of a polygon

Lemma 8. *Let $Q \subset \mathbb{H}^3$ be compact and right-angled, and let A be a k-gonal face. Then Q has at least $2k + 2$ faces. If it has exactly $2k + 2$, the unique face left after counting A, its k neighbors and its k perpendicular-edge opposites is disjoint from all k neighbors of A.*

Proof. Label the neighbors $A_1, \dots, A_k$ cyclically. At the vertex where A_i and A_{i+1} meet A, let B_i be the opposite face supplied by Lemma 7. Indices are cyclic. These are $1 + 2k$ distinct faces. Among the A_j, the only neighbors of B_i are A_i, A_{i+1}. The A_j themselves have just their cyclic adjacencies: any other adjacency, together with A, would give an actual vertex of A by Lemma 5.

Call any further face extra. Such a face meets at most one A_i. Indeed two nonconsecutive A_i cannot both meet it by Lemma 6; meeting two consecutive ones would force it to be the already counted B_i, by Lemma 7.

Pass to the dual triangulated two-sphere. Every vertex has degree at least five, since every face of Q is a compact right-angled polygon. We use the elementary ear fact: a triangulated polygon with no interior vertices has a vertex of degree two within that polygon. If one boundary edge is prescribed, such an ear can be chosen away from its endpoints. For example, root the tree dual to its triangles at that edge and take a leaf away from the root; for a single triangle take the third vertex.

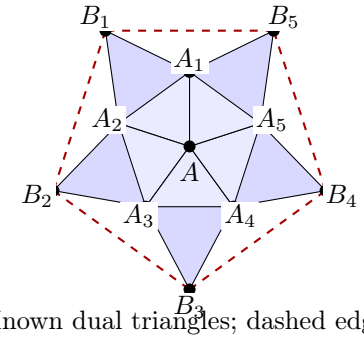
Suppose first there is no extra face. Each A_i has exactly the five known neighbors

$$A,\ A_{i-1},\ A_{i+1},\ B_{i-1},\ B_i.$$

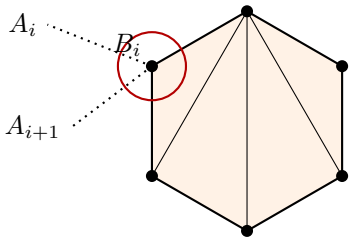
The cyclic link of A_i forces the edge $B_{i-1}B_i$. The triangles around A and the A_i form a disc whose complementary disc has boundary $B_1, \dots, B_k$ and no interior vertices. An ear in this disc has degree two there, and has only its two outside neighbors A_i, A_{i+1}. Its total degree would be four, a contradiction.

Now suppose there is exactly one extra face C, and it meets A_i. It meets no other A_j. The five-neighbor argument still forces $B_{j-1}B_j$ for $j \neq i$. At A_i its cyclic link instead has the segment B_{i-1}, C, B_i. The complementary disc is therefore a polygon with C inserted into the B-cycle, again without interior vertices. Every B_j has just two outside A-neighbors, so needs degree at least three within this disc. The vertex C has just the outside neighbor A_i, so needs degree at least four there. An ear is impossible. Hence the unique extra face meets none of the A_j, as asserted. □

The equality pattern can be recognized without a small-polyhedron list. When there is exactly one extra face and it meets none of the A_i, it is the unique interior vertex of the complementary B-disc. A chord between two B_i would cut off a polygon without this interior vertex. The ear fact, applied away from that chord's endpoints, again gives a B_i of total degree four. Thus there



Known dual triangles; dashed edges are forced when there is no extra face.



An ear has two neighbors in the disc and only two outside: total degree four.

Figure 1: The dual closing argument. The left panel depicts the cyclic incidences for $k = 5$; the argument works for every $k \geq 5$. The right panel is a schematic triangulated complementary disc. These are combinatorial drawings, not metric hyperbolic models.

are no chords and the extra face joins every B_i. The result has two k-gons and two cyclic rows of k pentagons: the Löbell face lattice. This agrees with the usual description in [5, p. 17 of the author's version]. The capacity argument uses only the proved disjointness of the extra face. The full face lattice will also be used, with an explicit metric-uniqueness input, in Section 7.

4 Capacity around a three-dimensional facet

Proof of Theorem 1. Let F have V vertices and M two-faces. Trivalence and Euler's formula inside F give

$$M = V/2 + 2. \tag{6}$$

There are M neighboring facets $F_1, \ldots, F_M$ and V distinct perpendicular-edge opposites R_x. Thus $1 + M + V$ facets are already counted. Call any other facet extra.

Suppose the ridge $F \cap F_i$ has k_i sides. Inside F_i, the known faces are precisely this ridge, its k_i neighbors coming from the other F_j, and its k_i perpendicular-edge opposites coming from the R_x at its vertices. The perpendicular edges in F_i are the same edges of P used in Lemma 7. If a different F_j meets F_i, their three pairwise orthogonal normals with that of F give an actual common edge by Lemma 5; hence it is already one of those k_i neighbors. The opposite-incidence equivalence in Lemma 7 makes the list of R_x exact as well.

Let e_i count the further faces of F_i, all coming from extra facets of P. Lemma 8 gives $e_i \geq 1$. An extra facet has footprint at most two among the F_j. Indeed its footprint is a clique by Lemma 6. Three members of this clique have mutually orthogonal normals in the Lorentz space $a^\perp$ of signature $(3, 1)$, where a is the normal of F. Together with a, they give an actual vertex x of F. A facet meeting all three is R_x, already counted. A larger orthogonal collection cannot fit in $a^\perp$.

There is an additional restriction when $e_i = 1$. The equality part of Lemma 8 says that its unique extra face is disjoint from all the k_i neighboring faces of $F \cap F_i$ inside F_i. The parent extra facet cannot also meet another F_j: such a meeting would force a common edge with F_i, F_j, contradicting that disjointness. This extra facet therefore has singleton footprint $\{i\}$.

Let s be the number of indices with $e_i = 1$. Their extra facets are distinct and cannot serve any other index. The other $M - s$ indices require at least $\sum_{e_i \geq 2} e_i$ incidences. Since every remaining extra facet serves at most two indices, the number of distinct extras is at least

$$s + \frac{1}{2} \sum_{e_i \geq 2} e_i \;\geq\; s + (M - s) = M.$$

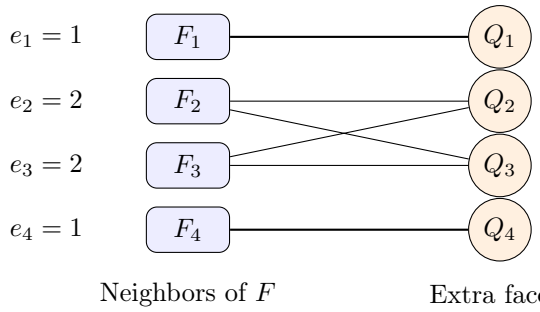


Figure 2: The final capacity count, shown as an incidence graph rather than a claimed polytope realization. An index with $e_i = 1$ consumes a distinct singleton. Every other index needs at least two incidences, while each remaining extra facet can serve at most two indices.

Adding the previously counted facets and using (6) proves

$$f \geq 1 + M + V + M = 2V + 5.$$

□

5 Pentagon excess and the volume reduction

Because P is simple, $e = 2v$. Euler's formula for its boundary gives $v - e + r - f = 0$, hence $r = v + f$. At each vertex the link is a tetrahedron, so six ridges meet there. Counting their corners yields

$$\sum_R k(R) = 6v.$$

It follows that

$$W = 6v - 5r = v - 5f \geq 0.$$

Inside a facet F, trivalence gives $2E_F = 3V$ and Euler gives $M = V/2 + 2$. Therefore

$$d(F) = 3V - 5M = \frac{V - 20}{2}. \tag{7}$$

Equivalently $d(F) = M - 12$. Each ridge belongs to two facets, so $\sum_F d(F) = 2W$. These identities also show that $d(F) \geq 0$.

For completeness the compact volume formula can be recovered from the reflection orbifold. The stabilizer of the interior of a codimension-j face is a product of j groups of order two. Its Euler characteristic is consequently

$$\chi_{\text{orb}} = 1 - \frac{f}{2} + \frac{r}{4} - \frac{e}{8} + \frac{v}{16} = \frac{v - 4f + 16}{16}.$$

Chern–Gauss–Bonnet in dimension four gives

$$\text{Vol}(P) = \frac{4\pi^2}{3}\chi_{\text{orb}} = \frac{\pi^2}{12}(v - 4f + 16) = \frac{\pi^2}{12}(f + W + 16),$$

as in (2).

Proof of Theorem 2. Let $F_1, \ldots, F_M$ be the distinct neighbors of F, and let $F \cap F_i$ have k_i sides. Lemma 8 inside F_i gives $d(F_i) \geq 2k_i + 2 - 12 = 2(k_i - 5)$. Since $\sum_i (k_i - 5) = d(F)$, the nonnegative excesses of these distinct facets satisfy

$$2W = \sum_Q d(Q) \geq d(F) + \sum_i d(F_i) \geq 3d(F).$$

Together with (7), integrality and Theorem 1, this proves (4). Shared ridges are allowed in the selected facet excesses: each is counted twice in $\sum_Q d(Q) = 2W$ as well. No disjointness of their ridge contributions is being assumed. □

Proof of Corollary 3. The regular right-angled 120-cell has $v = 600$, $f = 120$ and $W = 0$, hence volume $34\pi^2/3$. In its double across a dodecahedral facet, the 20 common vertices cease to be vertices and the 12 pairs of neighboring facets merge. Thus $v = 2 \cdot 600 - 2 \cdot 20 = 1160$, $f = 2 \cdot 120 - 2 - 12 = 226$ and $W = v - 5f = 30$, giving volume $68\pi^2/3$. The first volume comparison is therefore $f+W \leq 120$. A facet with $V = 48$ already forces $f+W \geq 101+21 = 122$ by Theorem 2. Since its lower bound increases with V and V is even, $V \leq 46$. Below the double, $f + W < 256$; $V = 98$ would force $f + W \geq 201 + 59 = 260$, so $V \leq 96$. The endpoints 46 and 96 are not excluded by these calculations. □

6 Metric inputs and the zero-excess case

We use the following two three-dimensional facts in the remaining arguments, rather than include them in the capacity proof. First, a compact right-angled three-polyhedron with 12 faces, equivalently 20 vertices, is the regular right-angled dodecahedron; see [6, Section 1.2]. Second, a fixed marked combinatorial type with fixed nonobtuse dihedral angles has at most one compact hyperbolic three-dimensional realization up to isometry, by the uniqueness clause of [7, Theorem 1.4]. The latter implies that every combinatorial symmetry preserving the angles is realized by an isometry. These are metric inputs; knowing only the face lattice of a dodecahedron or a Löbell polyhedron would not suffice.

Proposition 9. *$W = 0$ if and only if P is the regular right-angled 120-cell.*

Proof. The 120-cell has pentagonal ridges, so its excess vanishes. Conversely $W = 0$ makes every facet have $d(F) = 0$, hence 12 faces. By the stated three-dimensional classification every facet is a regular right-angled dodecahedron.

Let T be the regular right-angled 120-cell. Match one whole facet of P isometrically with a facet of T, with their interiors on the same inward side of its hyperplane. Take one of its pentagonal ridges R. The normal plane to the hyperbolic plane spanned by R is a positive two-space. Once the normal of the matched facet is fixed, there is a unique perpendicular normal line in it. Thus the facets adjacent across R in P and T have the same supporting hyperplane, with the same inward orientation.

Inside that hyperplane, a regular dodecahedron containing the fixed whole pentagon R is unique on either side of R. Any two placements differ by a symmetry of the regular pentagon, and those symmetries extend to the regular dodecahedron. The inward side of the first facet fixes the choice of side. Hence the adjacent whole facets agree.

Iterate across ridges. Connectedness of the facet graph of P shows that every facet of P matches a facet of T. The nonempty matched set is also closed under all neighbors in T, because every ridge of a matched whole facet is already a ridge in P. Connectedness of T's facet graph gives every facet of T. Their supporting halfspaces agree, so $P = T$. □

This uses regularity of *every* facet. It does not propagate from a single dodecahedral facet when its neighbors may be larger.

7 Regular pentagons and positive facets

Call a facet *positive* if $d(F) > 0$. Define two graphs on the positive facets. In G, an edge corresponds to a nonpentagonal ridge and has positive integral weight $k - 5$. In H, every adjacency of positive facets is an edge, including pentagonal shared ridges. Thus G is a spanning subgraph of H. Its weighted degree at F is $d(F)$ and its total edge weight is W.

Lemma 10. *Every positive facet has $d(F) \geq 2$. Graph G has no vertex of degree one. A degree-two vertex has equal edge weights $k-5$, its facet is a Löbell L_k, and $W \geq 5k-15$.*

Proof. A nonpentagonal ridge of size $k \geq 6$ gives $d(F) \geq 2(k-5) \geq 2$ by Lemma 8. More generally an edge of weight w gives $d(F) \geq 2w$ at either endpoint, excluding degree one. At degree two with weights w_1, w_2, $w_1 + w_2 \geq 2\max(w_1, w_2)$, so $w_1 = w_2 = w \geq 1$. Put $k = w + 5$. The facet has exactly $2k+2$ faces, hence the Löbell type proved after Lemma 8: two disjoint k-gonal bases and $2k$ lateral pentagons.

Its combinatorial k-fold rotation preserving both bases is realized by an isometry, using the uniqueness input in Section 6. The convex base is therefore a regular right-angled k-gon. If L_k is its edge length, the triangle from its center to a vertex to an adjacent edge midpoint has angles $\pi/k, \pi/4, \pi/2$, giving

$$\cosh(L_k/2) = \sqrt{2}\cos(\pi/k), \qquad \cosh L_k = 1 + 2\cos(2\pi/k). \tag{8}$$

Write $L = L_5$, so $\cosh L = \phi = (1+\sqrt{5})/2$. For $k \geq 6$, $L_k > L$. Each lateral pentagon contains a base edge of length L_k and is consequently not a regular right-angled pentagon. Its other facet neighbor cannot have zero excess, because every face of a regular dodecahedron is a regular pentagon of edge length L.

All $2k$ lateral neighbors are therefore positive with excess at least two. They and the two base neighbors are distinct: two convex facets cannot share two different ridges. Each base neighbor has excess at least $2(k-5)$. Including the original facet in the exact sum of facet excesses gives

$$2W \geq 2(k-5) + 2 \cdot 2(k-5) + (2k) \cdot 2 = 10k - 30, \qquad W \geq 5k - 15 \geq 15.$$

Only distinct facets were summed; their ridge contributions need not be disjoint. □

Lemma 11. *Every lateral pentagon of the right-angled Löbell L_6 has cyclic lengths (b, a, c, c, a), with*

$$\cosh b = 2, \qquad \cosh a = \sqrt{2}, \qquad \cosh c = \sqrt{3}.$$

Its edge on the hexagonal base is its unique longest edge.

Proof. Write the two bases as U, D and the two lateral rows as p_i, q_i, with indices modulo six. The neighbors of p_i are $U, p_{i-1}, p_{i+1}, q_{i-1}, q_i$; those of q_i are $D, q_{i-1}, q_{i+1}, p_i, p_{i+1}$. The combinatorial reflection $p_i \mapsto p_{-i}$, $q_i \mapsto q_{-i-1}$, fixing U, D, is realized isometrically by three-dimensional uniqueness. It fixes p_0 and exchanges its two p-neighbors and its two q-neighbors. Its lengths therefore have the stated order, with b on U. Rotation and exchange of rows cover all lateral faces.

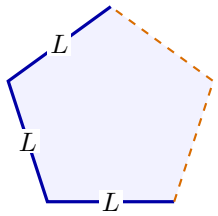
Equation (8) gives $\cosh b = 2$. For cyclic lengths s_i in a right-angled pentagon, the elementary identity is

$$\cosh(s_i) = \sinh(s_{i+2})\sinh(s_{i+3}).$$

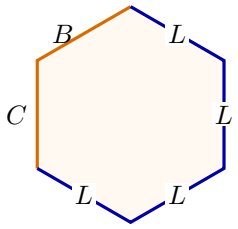
To derive it directly, put the common endpoint of sides $u = s_{i+2}$, $v = s_{i+3}$ at $(1,0,0)$ in the Lorentz plane, with their outgoing directions the two coordinate axes. The outward normals beyond their far ends are $(\sinh u, \cosh u, 0)$ and $(\sinh v, 0, \cosh v)$. Their product is $-\sinh u \sinh v$; their connecting perpendicular is side s_i, so it also equals $-\cosh(s_i)$. In the lateral pentagon this gives $2 = \sinh^2 c$ and $\cosh a = \sinh c \sinh a$. Hence $\cosh c = \sqrt{3}$, $\coth a = \sqrt{2}$ and $\cosh a = \sqrt{2}$. Since $2 > \sqrt{3} > \sqrt{2}$, the longest edge is unique. □

Proposition 12. *If G has a degree-two vertex, then $W \geq 20$.*

Proof. For $k \geq 7$ the preceding bound $W \geq 5k-15$ suffices. Suppose $k = 6$, and call the original facet F, its two base neighbors B_1, B_2, and their respective six lateral neighbors $A_{j,i}$, $1 \leq i \leq 6$. These fifteen facets are distinct and each has excess at least two, giving a baseline of thirty in $\sum d = 2W$.



Three fixed edges determine two orthogonal closing walls.



Two adjacent positive carriers leave a forbidden zero-carrier block.

Figure 3: The polygon obstruction. Blue edges have the regular-pentagon length L. In the right panel, a nonpentagonal ridge with only the two adjacent positive carriers B, C would have at least four consecutive blue edges. The drawings show the boundary order, not Euclidean representations of the hyperbolic right angles.

If $d(A_{j,i}) = 2$, then $A_{j,i}$ is also L_6: the edge-weight inequality in Lemma 10 forces two unit edges. Its shared pentagon with F is lateral in both facets. Its unique longest edge lies on B_j when seen in F, and on a hexagonal base when seen in $A_{j,i}$. These are the same edge. Exactly three parent facets contain its interior, namely $F, A_{j,i}, B_j$, by compact simplicity. The hexagonal-base parent in $A_{j,i}$ must therefore be B_j, since $F \cap A_{j,i}$ is pentagonal. Thus $A_{j,i}$ and B_j share a hexagon.

Let m_j count the six lateral neighbors in row j whose excess exceeds two. Their excess above baseline is at least m_j. The other $6 - m_j$ give that many distinct unit G-edges at B_j, in addition to FB_j. Consequently

$$d(B_j) - 2 \geq \max(0, 5 - m_j), \qquad m_j + \max(0, 5 - m_j) \geq 5.$$

The two rows and their bases are disjoint selected facet sets. Adding their extra excesses to the baseline gives $2W \geq 30 + 5 + 5 = 40$. Other positive facets only increase this sum; no disjointness of ridge contributions is assumed. □

Lemma 13. *A compact right-angled polygon with three consecutive edges of length L is the regular right-angled pentagon. In a nonpentagonal ridge of P, three consecutive edges cannot all be carried by zero-excess third facets.*

Proof. Match the inward-oriented three-edge chain to three edges of the regular right-angled pentagon. Fixed lengths and right-angle inward turns determine the chain uniquely. At either end, the next supporting line is the perpendicular through its endpoint. These two lines are therefore the remaining two lines of the regular pentagon, with the same inward orientations. They meet orthogonally. Crucially, they are actual walls of the original polygon. The projection in Lemma 5, now in $\mathbb{H}^2$, gives an actual finite vertex at their intersection: all other wall inequalities stay strict. Thus no further wall can truncate the putative closing vertex. The boundary closes along those two lines, and the polygon is the matched regular pentagon.

If an edge of a ridge $R = F \cap A$ is carried by a zero-excess third facet B, it is an entire edge of the regular pentagon $F \cap B$ in the regular dodecahedron B. Its length is L. Apply the first assertion to three such consecutive edges. □

For a positive facet F, let J_F be the graph induced by H on its neighbors. Its vertices are the positive facets meeting F. Call a vertex of J_F *eligible* if its degree is at least three, or if its degree is two and its two neighbors are not adjacent.

Lemma 14. *Every G-neighbor of F is eligible in J_F.*

Proof. For a nonpentagonal ridge $R = F \cap A$, the degree of A in J_F is exactly its number of positive third-facet edge carriers. A common neighbor gives an actual edge of R by projection of three orthogonal normals. If there are at most one such carrier, its other at least $k-1 \geq 5$ edges form a consecutive zero-carrier block, contradicting Lemma 13. If there are exactly two and they are adjacent in J_F, projection of four orthogonal normals gives an actual common vertex in R. Their edges are adjacent; the remaining $k-2 \geq 4$ consecutive edges are zero-carried, again impossible. □

Lemma 15. *A graph on at most four vertices with no K_4 and no induced C_4 has at most two eligible vertices.*

Proof. If it has a degree-three vertex, the remaining three vertices induce zero, one or two edges; three would give K_4. With zero or one, only the degree-three vertex can be eligible. With two, they form a path. Its middle and the original vertex have degree three, while each endpoint has two adjacent neighbors and is ineligible. If there is no degree-three vertex, all components are paths or cycles. Triangles have no eligible vertex, C_4 is excluded, and a path on at most four vertices has at most two internal vertices. □

Corollary 16. *If G has minimum degree at least three, then H has minimum degree at least five. In particular both conclusions hold when $0 < W < 20$.*

Proof. The assertion about G when $0 < W < 20$ follows from Lemma 10 and Proposition 12. Assume now that G has minimum degree at least three. If a positive facet F had at most four H-neighbors, J_F would have at least three eligible vertices by Lemma 14. It has no K_4, since together with F that would give five orthogonal unit normals in signature $(4, 1)$. It has no induced C_4: the two nonadjacent pairs in such a square would be orthogonal Lorentzian normal planes, giving two negative directions. Equivalently apply Lemma 6 to either disjoint pair. Thus Lemma 15 gives a contradiction. □

8 The small-graph obstruction

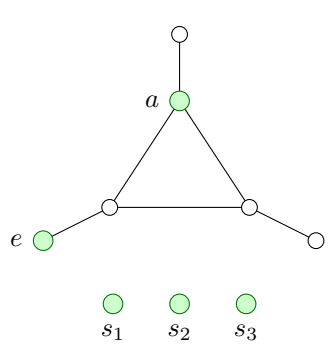
Lemma 17. *A graph C on nine vertices with maximum degree at most three and no induced pair of disjoint edges has an independent set of size five.*

Proof. Suppose first it contains a triangle T. Its vertices have only three degree slots outside T in total. Partition the other six vertices into D, those having a neighbor in T, and S, those having none. Then $|D| \leq 3$, $|S| \geq 3$. Set S is independent, because an edge in S with a triangle edge would induce two disjoint edges. If $|S| \geq 4$, add any triangle vertex.

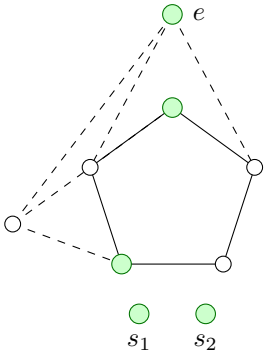
Otherwise $|S| = |D| = 3$, and all three slots are used, as a perfect matching from T to D. There is no D–S edge: such an edge and the triangle edge avoiding its D-endpoint's unique neighbor would induce two disjoint edges. Thus S consists of three isolated vertices. A triangle vertex and a D-vertex matched to a different triangle vertex are independent. With S they give five.

Suppose now C is trianglefree. If it is bipartite, one independent part has at least five vertices. Otherwise a shortest odd cycle is induced. Its length is five, since any induced cycle of length at least six contains an induced pair of disjoint edges. Write this cycle as $T = (t_0, \ldots, t_4)$.

Each cycle vertex has at most one outside neighbor, so there are at most five contacts to the other four vertices. An outside vertex has at most two cycle neighbors, nonadjacent by trianglefreeness. It cannot have exactly one, say t_i: its edge to t_i and the cycle edge $t_{i+2}t_{i+3}$ would induce two disjoint edges. Thus outside vertices have zero or two cycle neighbors, and at least two have zero. Each zero-contact vertex d is isolated. For if de were an outside edge, the at most two cycle neighbors of e cannot cover all cycle edges; an uncovered cycle edge with de is forbidden.



Triangle case: three isolated vertices and an unmatched pair give five.



C_5 case: two isolated vertices, one outside vertex and a cycle pair.

Figure 4: Independent sets in Lemma 17, highlighted in green. Only contacts used in the argument are shown. Additional edges among the unselected outside vertices do not affect the highlighted set; these are diagrams of the proof cases, not polytope realizations.

If there are at least three isolated vertices, add two independent cycle vertices. If exactly two, take another outside vertex e. Deleting its two nonadjacent cycle neighbors leaves three cycle vertices with an independent pair. That pair, e and the two isolated vertices form an independent five. □

Proposition 18. *If G has minimum degree at least three, there are at least ten positive facets.*

Proof. Let N be the number of positive facets, and suppose $N \leq 9$. Corollary 16 gives minimum degree five in H. Let C be the complement of H. It has no independent five, since H has no K_5, and no induced pair of disjoint edges, since H has no induced square. Its maximum degree is at most $N - 6$.

For $N \leq 5$, minimum degree five in H is impossible; $N = 6$ would give $H = K_6$. For $N = 7$, C has maximum degree one and at most one edge, since two would form a forbidden induced pair. It therefore has an independent set of size at least six.

For $N = 8$, maximum degree in C is two. At most one component can contain an edge. That component is a path on at most four vertices or a cycle on at most five: longer paths or cycles contain an induced pair of disjoint edges. The path cases, together with isolated vertices, give an independent set of size at least six. Cycles of lengths three, four and five give independent sets of sizes six, six and five respectively. Finally $N = 9$ is excluded by Lemma 17. All cases contradict the absence of an independent five, proving $N \geq 10$. □

The proof uses H, not just G, when excluding induced squares. A diagonal of G might be a pentagonal adjacency and hence present in H; it cannot be discarded from the normal geometry.

9 Five-neighbor links and the positive-excess gap

For the remaining small-excess cases we need the exceptional edge lengths, not only the number of positive carriers. We first record two elementary polygon facts.

Lemma 19. *A compact right-angled pentagon with two adjacent edges of length L is regular. In particular it is regular if any three edges have length L.*

Proof. Match the two-edge chain to the regular pentagon. This fixes its three vertices and the perpendicular supporting lines through its two outer endpoints. These lines are ultraparallel, as in the regular model. The remaining fifth side is their unique common perpendicular, fixing

the entire pentagon. Among any three edges of a pentagon two are adjacent, giving the second assertion. □

Lemma 20. *Suppose four specified edges of a compact right-angled hexagon have length L. Its other two side positions are opposite, and both remaining lengths are $2L$.*

Proof. Otherwise three of the four fixed positions are consecutive, contrary to Lemma 13. Write the cyclic lengths as a, L, L, d, L, L and put $A = \cosh a$, $D = \cosh d$, $p = \cosh L = \phi$. The right-angled hexagon relations are

$$A = \frac{p^2 + D}{p^2 - 1}, \qquad D = \frac{p^2 + A}{p^2 - 1}.$$

They also have a short direct Lorentz derivation. For cyclic sides a, b, c, d, e, f, let x, y be the future unit endpoints of a and n_a its outward normal. The outward normals of c, e are

$$n_c = \sinh(b)x - \cosh(b)n_a, \qquad n_e = \sinh(f)y - \cosh(f)n_a.$$

Their product is $\cosh(b)\cosh(f) - \cosh(a)\sinh(b)\sinh(f) = -\cosh(d)$: the connecting perpendicular is side d. This gives the first relation and, on exchanging opposite sides, the second. Since $p^2 - 1 = p$, subtraction gives $A = D$, and

$$A = D = \frac{p^2}{p - 1} = p^3 = \cosh(2L).$$

The lengths are positive, so both equal $2L$. □

Lemma 21. *If G has minimum degree at least three, a facet of excess three cannot have exactly five positive neighbors.*

Proof. Suppose $d(F) = 3$ and F has five positive neighbors. Its weighted G-degree and minimum unweighted degree force exactly three unit edges. It therefore has three hexagons and twelve pentagons. Put $J = J_F$. Its five vertices represent the positive neighbor faces of F; mark the three hexagons. Graph J has no K_4 and no induced C_4. Every other face of F borders a zero-excess facet, so is a regular pentagon. Thus every edge of a marked hexagon whose carrier is absent from J has length L. Actual edge and vertex intersections follow from Lemma 5; graph adjacencies are not merely formal.

A marked vertex has degree at least two. At degree two, Lemma 20 forces its neighbors to be nonadjacent and its two carrier edges to be opposite, both of length $2L$. At degree three, its neighbor graph cannot be a three-vertex path: that would place its three positive carrier edges consecutively in the hexagon, leaving three consecutive L-edges, contrary to Lemma 13. A triangle among its neighbors is already excluded by the absence of K_4. We now exhaust J.

If J has a universal vertex v, its other four vertices induce a trianglefree graph with no induced square, hence a forest B. A B-degree-zero vertex has J-degree one and is unmarked. A B-degree-one vertex has two adjacent J-neighbors and is unmarked. A B-degree-two vertex has three J-neighbors forming a path centered at v, and is unmarked. Only a B-degree-three vertex remains eligible for a mark, and a four-vertex forest has at most one. Including v gives at most two marks, a contradiction.

We may therefore assume J has maximum degree three. Suppose it contains a triangle T, with outside vertices d, e. If de is absent, an outside vertex with zero or one T-contact has degree at most one; with two contacts its degree-two neighbors are adjacent; three contacts give K_4. Neither outside vertex can be marked. A marked triangle vertex must have a unique outside contact, and that outside vertex must contact no other triangle vertex: otherwise its three neighbors form a path. There are only two outside vertices, so at most two marks.

If de is present, let $D, E \subset T$ be their contact sets. They are disjoint, since a common triangle neighbor would have degree four. They are nested, since $x \in D \setminus E$ and $y \in E \setminus D$ give an induced square x, d, e, y. Thus one set is empty; say $E = \varnothing$, and $|D| \leq 2$. If D is empty there are no marks. If $|D| = 1$, only d and its triangle neighbor can be marked. If $|D| = 2$, only d can be marked: its triangle neighbors have three-neighbor paths, while the remaining triangle vertex has two adjacent neighbors. In all cases e has degree one. Thus no graph containing a triangle has three marks.

Finally suppose J is trianglefree. It is a forest or C_5: any cycle has length five, and a chord would create a triangle. A forest on five vertices with at least three vertices of degree at least two must be P_5, with its three internal vertices marked. Each marked hexagon has degree two and its carrier edges have length $2L$. An endpoint of P_5 represents a pentagon, with four zero-carried L-edges. It is regular by Lemma 19, contradicting the shared $2L$-edge. In C_5 the three marked hexagons again have $2L$ carrier edges. One of the other two vertices is adjacent to a marked vertex. Its pentagon has three zero-carried L-edges, so is regular by the same lemma; its shared $2L$-edge is impossible. Every possible J has been excluded, proving the local assertion. □

The local proof needs the marked polygon geometry. The weaker abstract conditions on ten vertices alone are consistent: the graph $C_5[K_2]$, obtained by replacing each cycle vertex by an adjacent pair and joining consecutive pairs completely, has minimum degree five, no K_5 and no induced C_4. We have not excluded it by an unsupported small-graph assertion.

Lemma 22. *Let C have ten vertices, no induced pair of disjoint edges, and maximum degree at most four. If at most two vertices have degree four, then C has an independent set of size five.*

Proof. If at most one vertex has degree four, delete it, or any vertex if there is none. The remaining nine-vertex graph is subcubic, so Lemma 17 applies. If two degree-four vertices u, v are adjacent, deleting u again leaves a subcubic graph.

It remains to treat nonadjacent degree-four vertices u, v. Set

$$X = N(u) \cap N(v), \quad t = |X|, \quad A = N(u) \setminus N(v), \quad B = N(v) \setminus N(u),$$

and let S consist of the other vertices adjacent to neither u nor v. These partition the other eight vertices, with $|A| = |B| = 4 - t$ and $|S| = t$. All these vertices have degree at most three. Every $a \in A$ meets every $b \in B$, since otherwise ua, vb induce two disjoint edges. If A is nonempty, its vertices have degree at least $1 + (4 - t)$, so $t \geq 2$. Thus $t = 2, 3, 4$.

An edge yz in S must collectively contact every vertex of $N(u) \cup N(v)$: an uncovered $x \in N(u)$ would give the induced pair yz, ux, and similarly for v. Its endpoints have at most four spare degrees after yz. For $t = 2, 3$ the union has six or five vertices, respectively, so S is independent.

At $t = 2$, the two-element sets A, B are saturated by their complete cross contacts and their respective u, v contacts. Thus A is independent and has no S contacts. The set $\{v\} \cup A \cup S$ is an independent five. At $t = 3$, simply take $\{u, v\} \cup S$.

At $t = 4$, $A = B = \varnothing$. An S-edge yz must use all four spare degrees on the four vertices of X. Neither endpoint can have another S-neighbor. Any second S-edge would then induce two disjoint edges with yz. Hence S has at most one edge and an independent triple. Together with u, v this gives five. □

Proof of Theorem 4. The zero case is Proposition 9. Suppose $0 < W \leq 16$. By Proposition 12 and Lemma 10, graph G has minimum degree at least three. Let N be its number of vertices and E its unweighted edge count. Then

$$3N \leq 2E \leq 2W \leq 32.$$

Proposition 18 gives $N \geq 10$, so $N = 10$. Each positive facet has excess at least three, and the total excess is at most thirty-two. Thus at most two of them have excess greater than three. By

Lemma 21, every other facet has H-degree at least six; the exceptional two have degree at least five by Corollary 16.

The complement C of H therefore has maximum degree four, with at most two vertices of degree four. It has no induced pair of disjoint edges because H has no induced square. Lemma 22 gives an independent five in C, a forbidden K_5 in H. This contradiction excludes all $0 < W \leq 16$ at once, proving the theorem. □

10 Remaining volume questions

The results leave the inequality $f + W \geq 120$ unproved. A distinct polytope at or below the 120-cell volume must have $W \geq 17$ and at most 46 vertices in every facet, but these conditions do not classify its positive facets or their compatibility. Below the double the local cutoff is 96; no universal second-volume gap or equality classification follows.

In particular Theorem 1 starts at 45 facets for a dodecahedral facet. Recognition in Proposition 9 requires every facet to be dodecahedral. Neither statement proves that an arbitrary compact right-angled four-polytope is a union of regular 120-cells.

These are written geometric and graph-theoretic proofs. They use no numerical search or finite polytope census. Their external metric inputs are stated in Section 6; the normal, topological, graph and orbifold interfaces have not been formalized end to end in Lean. Source reviews and execution records are maintained separately; typesetting is not proof verification. No published priority claim is attached to these inequalities.

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